Open Source as a Research Method

From the research paper to billions of devices

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This morning, without knowing it…

2019	20 k+	10^9
FIRST RELEASE	GITHUB STARS	DEVICES

- You opened a web page: **simdutf** validated its text.
- Your app read some JSON: **simdjson** parsed it.
- A number went from text to binary: **fast_float** took care of it.

Node.js, Chrome, Safari, Rust, Go, .NET, ClickHouse… code written in a university lab, in Quebec.

The question

How does academic research end up
in the pockets of billions of people?

Where do ideas come from?

> *I never come up with anything by going off to a mountaintop to think. Instead, my ideas come from two sources: talking to real users with real problems and then trying to solve them. This ensures I come up with ideas that somebody cares about and the rubber meets the road and not the sky.*

Michael Stonebraker — 2014 Turing Award · Ingres, PostgreSQL, Vertica

The linear model of innovation

- The professor thinks. The professor publishes.
- The engineer reads, designs.
- The customer consumes.

Toward a real model of innovation

- Industrial revolution: underwear.
- Where the keyboard came from.
- ChatGPT: GPUs (games), text (the web), chat (Discord).
- Relativity (trains)

Part 1

Open source as a research *method*

> *Open source is not just a way to publish research results: it is a research method in its own right.*

1. Reproducibility

- The code **and** the benchmarks are public.
- Anyone can rerun the measurements — on their own machine, with their own data.
- So anyone can **prove you wrong**. That is the point.

A performance gain that does not survive a stranger's machine was never a result.

2. Peer review that is actually real

JOURNAL REFEREES

- 2 to 3 reviewers
- They read the description
- One verdict, once
- Anonymous, no follow-up

PUBLIC CODE REVIEW

- Hundreds of eyes
- They run the code
- A verdict on **every** release
- Signed, and you must answer

Code reviews, issues and pull requests are often a more demanding form of review than a journal's referees.

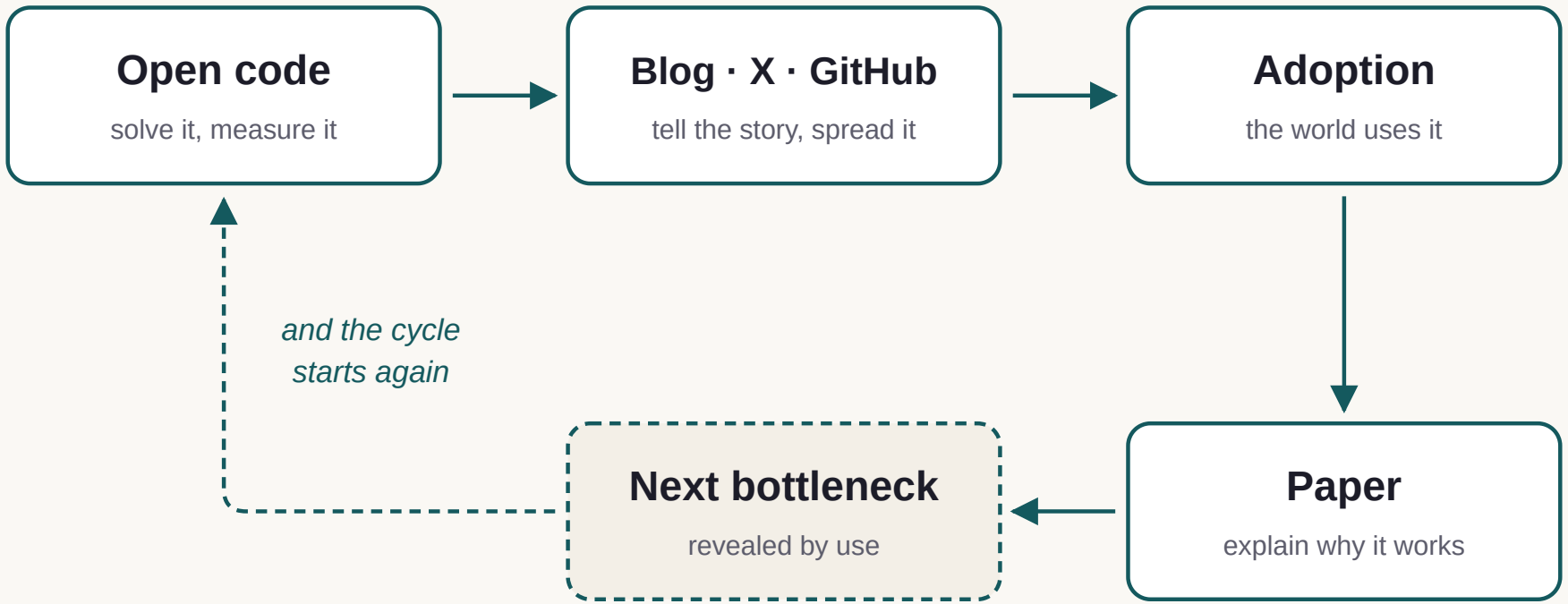
3. Real-world data comes to you

- **Adversarial cases**: it doesn't work, it isn't fast enough.
- **Exotic architectures**: ARM, POWER, RISC-V, WebAssembly.
- **Real workloads**: the files nobody would have thought to make up.

Part 2

A story: *simdjson*

The cycle


Open code
solve it, measure it
Blog · X · GitHub
tell the story, spread it
Adoption
the world uses it
Paper
explain why it works
Next bottleneck
revealed by use
and the cycle
starts again

Act 1 — parsing JSON

2019

- JSON is everywhere, and parsing it is **slow**: hundreds of megabytes per second, at best.
- **Code first**: simdjson, written with Geoff Langdale — several **gigabytes per second**, thanks to SIMD instructions.
- **Then a blog post**, GitHub, a few posts on X. Within days, thousands of readers.
- **Then adoption**: databases, analytics engines, JavaScript runtimes.
- **The paper comes after**: *Parsing Gigabytes of JSON per Second* (VLDB Journal).

The order matters: the software was already adopted when the paper came out — and the paper would not exist without the measurements that real use made possible.

Act 2 — an unexpected bottleneck: numbers

REVEALED BY USE

- Once everything else is fast, parsing **floating-point numbers** becomes the bottleneck.
- Nobody would have posed this problem *a priori*: profiling a real parser pointed to it.
- Result: the **Eisel-Lemire** algorithm, exact and fast (*Number Parsing at a Gigabyte per Second*).
- Adoption: Rust, Go, .NET, C++ — the standard libraries themselves.

Act 3 — another bottleneck: UTF-8

REVEALED BY USE

- Parsing JSON also means **validating** UTF-8: security depends on it.
- Goal reached: validation in **less than one instruction per byte**.
- That piece broke off to become a library of its own: **simdutf**.
- Adoption: Node.js, Bun, Deno, browsers — Unicode validation and transcoding for the web.

The same cycle, three times

PROBLEM	CODE	ADOPTION	PAPER
JSON parsing	simdjson	databases, JS engines	VLDB Journal
Number parsing	fast_float	Rust, Go, .NET, C++	Software: P&E
UTF-8 validation	simdutf	Node.js, Bun, browsers	Software: P&E

Each row grew out of the one before. None of them was in the original research plan.

Part 3

The honest tensions

Maintenance is thankless

- Success is paid for in **bug reports**, porting requests, questions.
- A security flaw in your code is a security flaw in Node.js.
- This work is **endless** and **invisible** — it shows up on no academic CV.

Publishing a paper means you are done. Publishing a library means you are just getting started.

The incentives are misaligned

WHAT COMMITTEES COUNT

- Papers
- Citations
- Grants

WHAT CHANGES THE WORLD

- Pull requests
- Maintained releases
- Users

There is no "adopted by a billion devices" box on a promotion form.

My takeaway

Open code **raises** the research questions,
then it **checks** the answers.

The lesson of technology transfer

> *The problem in this business isn't to keep people from stealing your ideas; it's making them steal your ideas.*

Howard Aiken — quoted by David Patterson, 2017 Turing Award

Thank you

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simdjson · fast_float · simdutf · Roaring Bitmaps